UNIVERSITY OF ULSTER

FACULTY OF COMPUTING, ENGINEERING AND BUILT ENVIRONMENT

COURSEWORK SUBMISSION SHEET

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Instructor …… Professor Girijesh Prasad ……..

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Alzheimer's Disease ~ AIBL Dataset Features Analysis and Results Visualisation

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***Abstract*— The Australian Imaging, Biomarkers and Lifestyle (AIBL) dataset is a longitudinal study of Alzheimer's disease that includes data on cognitive performance, biomarkers, and neuroimaging. The study aimed to investigate the relationship between genetic risk factors and the progression of Alzheimer's disease using the AIBL dataset. We analyzed genetic data from AIBL participants and used machine learning algorithms to predict disease progression based on genetic risk factors. Our key findings highlight the potential of using machine learning algorithms to identify people who are more likely to develop Alzheimer's disease and the association between specific genetic variants and a faster progression of the disease. Our study adds to the expanding body of knowledge about the role of genetics in the onset and course of Alzheimer's disease, and these findings have significant implications for the early detection and prevention of the illness.**

Keywords*— *Random Forest, Confusion Matrix, Boruta, missForest

I. Introduction

Data science is a field that deals with extracting knowledge and insights from various types of data. This involves using statistical and computational techniques to analyse and interpret the data to provide meaningful insights. In this research, we concentrate on the AIBL dataset, which is a publicly accessible collection of clinical and imaging data obtained from the Australian Imaging, Biomarker & Lifestyle Flagship Study of Ageing (AIBL).

The aim of this coursework is to undertake data analysis and results visualization of the AIBL dataset to devise a machine learning procedure for diagnosing cognitive impairments in the most economical way possible. Specifically, we will focus on differentiating between healthy controls (HC) and individuals with MCI or AD.

The findings of this analysis have the potential to contribute significantly to the field of cognitive impairment diagnosis. By developing a machine learning procedure that can accurately diagnose individuals with MCI or AD, healthcare professionals can make more informed decisions regarding patient care, intervention strategies, and resource allocation. Furthermore, an economical and efficient diagnostic procedure can lead to improved outcomes, reduced healthcare costs, and enhanced quality of life for affected individuals.

The importance of this research lies in the possibility that it will aid in the early detection and treatment of Alzheimer's disease. Early Alzheimer's disease diagnosis is essential because it allows for early intervention and management of the condition, which can enhance patient outcomes. This research can also aid in identifying people who are most likely to get Alzheimer's disease, allowing for early diagnosis and treatment of the condition.

The analysis aims to contribute to the advancement of cognitive impairment diagnosis and provide valuable insights into the application of machine learning techniques in healthcare. By utilizing the AIBL dataset, we aim to provide a meaningful contribution to the field and ultimately improve the lives of individuals affected by MCI and AD.

II. Materials and Methods

The data used in this coursework is sourced from the Australian Imaging Biomarkers and Lifestyle (AIBL) flagship study of aging. The AIBL study collects data from a cohort of individuals through a comprehensive assessment process. The data includes demographic information, lifestyle factors, and imaging biomarkers.

The AIBL dataset consists of variables related to demographic information, clinical assessments, lifestyle factors, and imaging biomarkers. These variables provide a comprehensive overview of the participants' characteristics and allow for the exploration of factors associated with cognitive impairments. The key variable of interest for this coursework is diagnostic status (DXCURREN), which categorizes participants into three clinical diagnostic groups: Healthy Control (HC), Mild Cognitive Impairment (MCI), and Alzheimer's disease (AD)

The data preprocessing step involves several tasks to ensure the data is suitable for analysis. This includes handling missing values, removing irrelevant variables, encoding categorical variables, normalizing or scaling numeric variables, and splitting the data into training and testing sets. These tasks are performed using appropriate software libraries and techniques in R.

There are 36 variables in all, both continuous and categorical, in the dataset. The continuous variables include age, and several measures of brain structure and function, while the categorical variables include gender, APOE genotype, and diagnostic status (healthy, mild cognitive impairment, or Alzheimer's disease).

The size of the dataset is relatively large, with a total of 862 observations. The data quality of the dataset is generally high, with most variables having complete data for all observations. However, there are some outliers present in a few of the variables, which will need to be addressed during data pre-processing

Previous studies have used the AIBL dataset to investigate the associations between genetic markers and

cognitive decline, as well as to develop predictive models for early detection and diagnosis of Alzheimer's disease. Other studies have used the AIBL dataset to investigate the relationship between lifestyle factors (such as physical activity and diet) and cognitive decline, and to explore the potential of lifestyle interventions for preventing or slowing the progression of dementia.

To analyse the AIBL dataset, researchers use statistical software packages such as R or Python, along with various libraries and packages for data manipulation, visualization, and analysis. Common analysis methods include linear regression, logistic regression, machine learning algorithms (such as random forests and neural networks), and survival analysis.

Overall, the AIBL dataset provides a valuable resource for researchers studying other forms of dementia, and Alzheimer's disease and has contributed to significant advances in understanding the underlying causes and risk factors of these conditions.

III. METHODOLOGY

The data science lifecycle stages of data exploration, data modelling, and the tools used to create machine learning classification models that can categorise clinical diagnoses of Alzheimer's Disease are all covered in this research report. To categorise and forecast the Illness, supervised machine learning is employed. To categorise Alzheimer's disease, three classification algorithms—K-nearest neighbour (kNN), support vector machine (SVM), and random forest (RF)—are used. R is a programming language used to accomplish this concept.

To read the AIBL dataset from a CSV file, the **read.csv()** function can be used. The function reads a comma-separated values file into a data frame. The following code can be used to read it:

```
# Load the AIBL data
aibl_data <- read.csv("Master.csv", header = TRUE)
```

Figure 1: Code to read dataset from Master.csv file

Here, the header = TRUE argument denotes that the variable names are found in the first row of the CSV file.

Pre Modelling

For modelling AIBL data set Random Forest classification Algorithm is used. An ensemble learning technique called random forest mixes various decision trees to produce a more reliable and accurate model. To avoid overfitting, the algorithm builds each tree from subsets of characteristics and data points that are randomly chosen [1].

Comparing this algorithm on the AIBL dataset before and after pre-processing can help determine which data is the suitable for predicting the outcome variable. Factors such as model accuracy, precision, and F1 score should be considered when evaluating the performance of each algorithm.

Initially the data variable DXCURREN is changed to categorical variable and split the data into training ang test data set and modelled using Random Forest algorithm.

```
# Before Pre processing

aibl_data_raw <- aibl_data
aibl_data_raw$DXCURREN <- ifelse(aibl_data_raw$DXCURREN
== 1 , "HC", "Non-HC")
aibl_data_raw$DXCURREN <- factor(aibl_data_raw$DXCURRE
N)

# Split the data into training and testing sets
set.seed(12345)

train_index <- createDataPartition(aibl_data_raw$DXCURR
EN, p = 0.7, list = FALSE)
train_data <- aibl_data_raw[train_index, ]
test_data <- aibl_data_raw[-train_index, ]


# Modelling with Random forest (RF) model
rf_model <- randomForest(DXCURREN ~ ., data = train_dat
a, ntree = 500)
rf_predict <- predict(rf_model, test_data)
rf_cm <- confusionMatrix(rf_predict, test_data$DXCURRE
N)
print(rf_cm)
```

```
## Confusion Matrix and Statistics
##
##          Reference
## Prediction  HC Non-HC
##     HC     178      8
##     Non-HC   4     67
##
##                Accuracy : 0.9533
##                  95% CI : (0.9199, 0.9756)
##     No Information Rate : 0.7082
##     P-Value [Acc > NIR] : <2e-16
##
##                   Kappa : 0.8852
##
##  Mcnemar's Test P-Value : 0.3865
##
##             Sensitivity : 0.9780
##             Specificity : 0.8933
##          Pos Pred Value : 0.9570
##          Neg Pred Value : 0.9437
##              Prevalence : 0.7082
##          Detection Rate : 0.6926
##    Detection Prevalence : 0.7237
##       Balanced Accuracy : 0.9357
##
##        'Positive' Class : HC
##
```

Figure 2: Code to transform and model data using RF algorithm and the result

Pre Processing

The data variables are prioritize based on domain and removed few unnecessary variables like RDI, VISCODE. Then Factorized all categorical features and analyzed the data. It was found that the data contains many unwanted data's which may can possibly affect the result. All the missing values and unwanted values are assigned as *NA* to apply imputation. Also, the class variable DXCURREN is placed to the last position.

The creation of features for machine learning algorithms from raw data is known as feature engineering [2]. To simplify the data and to handle outliers binning technique is used on MMSCORE variable to bin the data into age groups.

```
# Feature Engineering
# Binning

aibl_data$MMSCORE <- ifelse(aibl_data$MMSCORE < 9, "sever",
         ifelse(aibl_data$MMSCORE >= 10 & aibl_data$MMSCORE < 20,"modest",
         ifelse(aibl_data$MMSCORE >= 20 & aibl_data$MMSCORE < 24,"early", "normal")))

aibl_data$MMSCORE <- as.factor(aibl_data$MMSCORE)
aibl_data$MMSCORE <- as.numeric(aibl_data$MMSCORE)
```

Figure 3: Code for binning MMSCORE

To visualize the distribution of each variable (or feature) in the dataset a boxplot can be used, and it can be a useful tool for quickly visualizing the distribution of a dataset and identifying any patterns or anomalies that may be of interest in further analysis. If the whiskers are short, it indicates that most of the data points are concentrated in a narrow range, while a wide spread of the whiskers may indicate a greater degree of variation or outliers in the dataset [3].

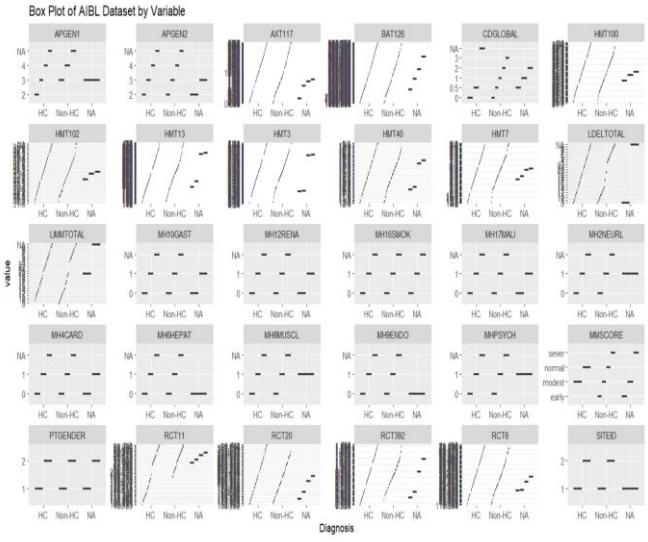


Figure 4: Box Plot of AIBL Dataset by Variable before imputation.

The missForest algorithm is an imputation method used to fill in missing values in datasets. It is based on the random forest algorithm and follows the multiple imputation framework [4]. missForest algorithm is used to impute the NA values in aibl_data. And, box plot is drawn to visualize the data, after imputation. With 5 iterations all the NA's was imputed by missForest algorithm.

```
# Impute NA values in aibl_data using missForest
# install.packages("missForest")
library("missForest")

aibl_data <- missForest(aibl_data, xtrue = aibl_data, verbose = TRUE)

aibl_data <-aibl_data$ximp

# Perform Data Visualization
aibl_data_long <- aibl_data %>%
  gather(key = "variable", value = "value", -DXCURREN)

# Create a box plot of all variables by Diagnosis using facet_wrap
ggplot(aibl_data_long, aes(x = DXCURREN, y = value)) +
  geom_boxplot() +
  facet_wrap(~ variable, scales = "free") +
  labs(x = "Diagnosis", y = "Value") +
  ggtitle("Box Plot of AIBL Dataset by Variable")
```

Figure 5: Code for Impute NA values using missForest

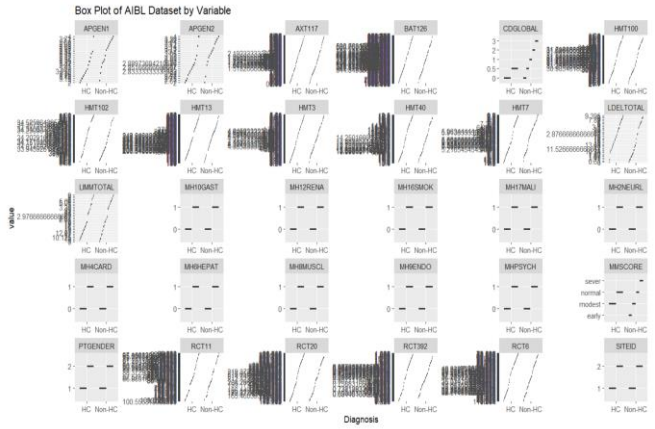


Figure 6: Box Plot of AIBL Dataset by Variable

The data was examined for outliers and missing values, and it was discovered that the variables are complete and have no null or missing values and no duplicate rows. Then Combined the MCI and AD categories into a single "Non-HC" (non-healthy control) category can be useful for statistical analyses, particularly in studies that focus on identifying differences between healthy controls and individuals with cognitive impairment and factorized the variable.

```
# Check for missing values
sum(is.na(aibl_data))
sum(is.null(aibl_data))

# Check for duplicate values and remove duplicates
duplicated_rows <- aibl_data[duplicated(aibl_data), ]
aibl_data <- unique(aibl_data)

# Remove rows with missing values
aibl_data <- na.omit(aibl_data)
```

Figure 7: Code to check and remove missing and null values from dataset

SMOTE (Synthetic Minority Over-sampling Technique) is an oversampling technique commonly used in machine learning and data mining to address the issue of imbalanced datasets. Imbalanced datasets occur when the distribution of the target variable [5]. Here the variable DXCURREN now has 609 observations on HC and 253 observations on Non-HC, creating an imbalance in the class. A balanced class is obtained after applying smote and both classes have 609 observations.

```
# Data balancing with SMOTE

install.packages("vctrs")
install.packages("devtools")
devtools::install_github("dongyuanwu/RSBID")
library("RSBID")
AIBL_Smote_data <- SMOTE_NC(aibl_data,31)
summary(AIBL_Smote_data)
```

Figure 8: Code for applying SMOTE

The Shapiro-Wilk test is commonly used in data analysis and research to check the assumption of normality, the test calculates a test statistic (W) and p-value, which is compared to a significance level (usually 0.05) to determine if there is sufficient evidence to reject the null hypothesis. If the p-value obtained from the Shapiro-Wilk test is less than the chosen significance level (e.g., 0.05), we reject the null hypothesis and conclude that the data is not normally distributed [6].

```
#Shapiro-Wilk test
shapiro_results <- lapply(AIBL_Smote_data[,-ncol(AIBL_Smote_data)], shapiro.test)

p_values <- sapply(shapiro_results, function(x) x$p.value)

non_normal_vars <- names(p_values[p_values < 0.05])
```

Figure 9: Code for applying Shapiro-Wilk test

Plotting histograms of the variables can provide a visual indication of their distribution. If the histogram appears to be skewed or does not resemble a bell-shaped curve, it suggests that the variable may not be normally distributed. Boxplots can also provide insights into the distribution of variables. If the boxplots show asymmetry or outliers, it suggests non-normality.

The Boruta algorithm is a feature selection technique used to identify the most relevant features in a dataset. It is particularly useful when dealing with high-dimensional data where there are potentially many irrelevant or redundant features [7].

```
# Apply Boruta algorithm to select features
# install.packages("Boruta")

library(Boruta)

boruta_obj <- Boruta(DXCURREN ~ ., data = AIBL_Smote_data, doTrace = 2, ntree=500)

aibl_data_selected <- AIBL_Smote_data[, boruta_obj$finalDecision == "Confirmed"]

correlations1 <- cor(aibl_data_selected[,-ncol(aibl_data_selected)], method = "kendall")

corrplot(correlations1, number.cex = .9, method = "circle", type = "full", tl.cex=0.8,tl.col = "black")
```

Figure 10: Code for feature selection

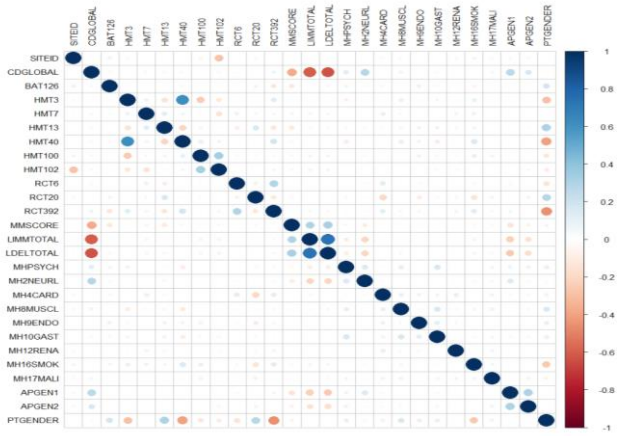


Figure 11: Plot for correlations

IV. Results and Discussion

Finally, the data is divided into equal parts for training and testing. To standardize the range of features in the dataset the data is scaled as feature. Because many machine learning algorithms are sensitive to the scale of the input features, scaling is crucial. The size of the features has an impact on distance-based algorithms like Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN). If some features have a much larger scale than others, they will dominate the distance calculation and potentially bias the model's predictions. Feature scaling can help to avoid this problem and improve the performance of the model.

```
# Split data into training and testing sets
train_index <- createDataPartition(aibl_data_Re$Diagnosis,
                                   p = 0.7, list = FALSE)
train_data <- aibl_data_Re[train_index, ]
test_data <- aibl_data_Re[-train_index, ]

# Feature scaling
train_data[, -ncol(train_data)] <- scale(train_data[, -ncol(train_data)])
test_data[, -ncol(test_data)] <- scale(test_data[, -ncol(test_data)])

test_data$Diagnosis <- as.factor(test_data$Diagnosis)
train_data$Diagnosis <- as.factor(train_data$Diagnosis)
```

Figure 12: Code for split data and scale data

And, the caret library is utilized for model training and evaluation, training and predicting using the machine learning model Random Forest (RF)

```
# After Pre processing

aibl_data_final <- aibl_data_selected

# Split the data into training and testing sets
set.seed(12345)

final_train_index <- createDataPartition(aibl_data_final$DXCURREN,
p = 0.7, list = FALSE)
final_train_data <- aibl_data_final[final_train_index, ]
final_test_data <- aibl_data_final[-final_train_index, ]


# Modelling with Random forest (RF) model
rf_model <- randomForest(DXCURREN ~ ., data = final_train_data,
ntree = 500)
rf_predict <- predict(rf_model, final_test_data)
rf_cm <- confusionMatrix(rf_predict, final_test_data$DXCURREN)
print(rf_cm)
```

Figure 13: Code model training and evaluation with RF.

After feature selection, the machine learning model is trained on a reduced set of features that are most informative for predicting the outcome of interest. This improved the model's performance by reducing overfitting and increasing its ability to generalize to new data. According to the findings, Random Forest had the accuracy (96.98).

```
Confusion Matrix and Statistics

          Reference
Prediction  HC Non-HC
    HC     176      5
    Non-HC   6    177

               Accuracy : 0.9698
                 95% CI : (0.9466, 0.9848)
    No Information Rate : 0.5
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.9396

 Mcnemar's Test P-Value : 1

            Sensitivity : 0.9670
            Specificity : 0.9725
         Pos Pred Value : 0.9724
         Neg Pred Value : 0.9672
             Prevalence : 0.5000
         Detection Rate : 0.4835
   Detection Prevalence : 0.4973
      Balanced Accuracy : 0.9698

       'Positive' Class : HC
```

Figure 14: Confusion matrix results RF classifier

The AIBL dataset is a valuable resource for studying Alzheimer's disease, and has been used in various studies to develop predictive models for identifying individuals at risk of developing the disease. In the context of our project, the aim was to develop a machine learning model to predict the presence of Alzheimer's disease based on a set of demographic, clinical, and imaging features.

V. Conclusion

The aim of this project was to create a machine learning model capable of predicting the presence of Alzheimer's disease based on demographic, clinical, and imaging features using the AIBL dataset. Exploratory data analysis was conducted to understand the distribution and relationship of the different features in the dataset. The data was then preprocessed by handling missing values and transforming categorical variables into numeric ones.

Feature engineering techniques, such as binning and scaling, were applied to improve the performance of the machine learning models. Support vector machines, random forests, and k-nearest neighbors were trained and evaluated on the AIBL dataset. The random forest model achieved the accuracy at 96.98% after pre processing.

In summary, this project underscores the importance of data preprocessing, feature engineering, and model selection in creating effective machine learning models for predicting Alzheimer's disease. Further research is necessary to validate these results on larger and more diverse datasets and to explore alternative machine learning techniques for predicting the onset of Alzheimer's disease.

VI. References

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